

SETLKW will wait till the lock is released, if it's held by another process, but *SETLKW* will return with an error if the lock is not available. *F_GETLK* will just check if the lock is available.

File-locking implementations: *flock*

The usage of *flock* is simpler than *fcntl*. On the flip side, *flock* does not have many features that a developer needs. Use *flock* just to lock or unlock, by passing a suitable operation flag. The *flock* command manages locks within shell scripts, or at the command line. The *flock* system call is used to apply or remove an advisory lock on an open file. Information on the *flock* system call is provided below.

Syntax:

```
int flock (int fd, int operation);
```

Available operations:

- *LOCK_SH* (shared lock, more than one process can hold)
- *LOCK_NB* (don't block when locking)
- *LOCK_EX* (exclusive lock; only one process at a time)

- *LOCK_UN* (remove an existing lock)

Examples of file-locking implementations of both *fcntl* and *flock* types are shown in Figure 13.

To sum up, let's look at the procedure for lock usage:

1. Declare a lock
2. Initialise the lock
3. Lock the critical section
4. Unlock the critical section

In this article, we have looked at how a file is accessed on the Linux platform, and its complete sequence. We showed different file-related system calls, and discussed how they work. We explained different file-synchronisation mechanisms available on Linux, and the procedures to access different kinds of files. This information is valuable to software architects who create entire software products on the Linux platform. **END** 

Acknowledgement

The authors would like to acknowledge Krishna Sudhakaran for his valuable assistance in the preparation of the view graphs.

References

1. Kernel Corner: *Starting with Linux Device Drivers*, by Dr B. Thangaraju, *LINUX For You*, May 2003, p 83
2. Linux supported file systems: <http://www.linux-tutorial.info/modules.php?name=MCContent&pageid=243>
3. List of file systems: http://en.wikipedia.org/wiki/List_of_file_systems
4. A comparison of file systems: http://en.wikipedia.org/wiki/Comparison_of_file_systems
5. Ext4 fs features: <http://kernelnewbies.org/Ext4>
6. Understanding the Linux Kernel, (3rd Edition) by Daniel P. Bovet, Marco Cesati, O'Reilly Publications, 2005
7. extundelete details: <http://extundelete.sourceforge.net/>
8. gparted documentation: <http://gparted.sourceforge.net/documentation.php>
9. Kernel Corner: *Examining Process Information*, by Dr B. Thangaraju, *LINUX For You*, January 2004, pgs 84-87
10. Kernel Corner: *Interfacing the proc file system with a kernel module*, by Dr B. Thangaraju, *LINUX For You*, February 2004, pgs 90-92
11. Kernel Corner: *Linux Kernel Locking Mechanisms for Kernel Programming*, by Dr B. Thangaraju, *LINUX For You*, September 2003, pgs 81-83;
12. Basics of System V Semaphore, by Dr B. Thangaraju, *LINUX For You*, July 2006, pgs 86-89;
13. The Intricacies of System V Semaphore, by V. Shobana and Dr B. Thangaraju, *LINUX For You*, April 2009, pgs 40-43.

By: Gururajan Narasimhan Erode and Dr B. Thangaraju

Gururajan Narasimhan Erode currently works with Wipro Technologies as a lead consultant in the Wireless and Embedded Domain. He has 16 years of experience after a masters degree in engineering, and has worked with various multinational companies in the USA, Germany and India. Gururajan's specialisation is in wireless communication, algorithms and data structures, and Linux systems programming. He can be reached at gururajan.erode@wipro.com.

Dr B. Thangaraju received his Ph.D. in Physics and worked as a research associate in the Indian Institute of Science from 1996 to 2001. From 2001, he has been working at Wipro Technologies as a senior consultant, and his core expertise is in the Linux kernel and embedded and real-time Linux. He has published more than 35 papers on Linux in renowned international and national journals, and has presented more than 20 technical papers at national and international conferences. He can be reached at balat.raju@wipro.com.

A Business Minded Computer Magazine

To BUY or SUBSCRIBE, please visit www.benefitmag.com



www.benefitmag.com